

Prime Pancakes (pancakes)

Stefan just bought a pancake shop, and he's going to sell N different kinds of pancake there. Each pancake is labeled with an integer number L_i . The label is a positive integer up to 6 digits long ($1 \leq L_i \leq 999\,999$). Different kinds of pancake can sometimes have the same label.



In a few hours the store will be open for business, but Stefan just remembered that he still hasn't decided the prices of each pancake type! In a hurry, Stefan decided to use his imagination and came up with a really interesting algorithm for deciding the price of a pancake.

Since Stefan loves prime numbers, the price is going to depend on “how prime” a pancake is:

- The starting price of a pancake is B euro.
- If the label's value L_i is prime, the price is increased by L_i + a “primeness bonus” of P euro.
- For each digit d of the label: if d is prime (that is, if it's either 2, 3, 5, or 7) then the price of the pancake will be increased by a corresponding fixed amount (D_2 , D_3 , D_5 , or D_7 euro).
- If the sum of all digits $\sum d$ of the label is prime, the pancake's price is increased by $\sum d$ euro.
- If the product of all digits $\prod d$ of the label is prime, the pancake's price is increased by $\prod d$ euro.

Even though this algorithm is quite convoluted, sadly it doesn't always yield prices that are high enough to make a profit. Noticing this, Stefan decided that it's okay to change the pancakes' labels slightly, to increase their price.

You will be allowed to change **at most one digit** from each label, but on one condition: the number of digits should stay the same, so you can't change the most significant digit to a zero, because that would reduce the total number of digits.

Help Stefan compute the **sum of all pancakes' prices** after changing up to one digit from each label!

📎 Among the attachments of this task you may find a template file `pancakes.*` with a sample incomplete implementation.

Input

The first line contains the only integer N . The second line contains N integers L_i . The third line contains two integers B and P . The fourth line contains the four values D_2 , D_3 , D_5 , and D_7 .

Output

You need to write a single line with an integer: the sum of all pancakes' prices after changing up to one digit from each label.

Constraints

- $1 \leq N \leq 100\,000$.
- $1 \leq L_i \leq 999\,999$ for each $i = 0 \dots N - 1$.
- $1 \leq B, P \leq 1\,000\,000$.
- $1 \leq D_2, D_3, D_5, D_7 \leq 1\,000\,000$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.
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- **Subtask 2** (30 points) $N \leq 1000$.
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- **Subtask 3** (70 points) No additional limitations.
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Examples

input	output
4 12 89 941 101 5 7 8 5 3 9	1907

Explanation

In the **first sample case**, we can do these changes:

- First label: $12 \rightarrow \mathbf{17}$, for a price of: 45.
- Second label: stays **89**, for a price of: 118.
- Third label: $941 \rightarrow \mathbf{991}$, for a price of: 1022.
- Fourth label: $101 \rightarrow \mathbf{701}$, for a price of: 722.

The sum of all prices, after the changes are done, is 1907.